

REPOSITORY HARDENING REPORT

512 Repository · Evidence-Sidecar (CVS) Repository
GAP Attribution Hardening Pass — Terminology Correction Across Both Repos

Date	17 April 2026
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Pass type	Terminology hardening — GAP attribution correction
Repos affected	2 — 512 (JonathanMastersWatson/512) and Evidence-Sidecar (JonathanMastersWatson/Evidence-Sidecar)
512 files changed	7 (6 corrected + 1 new CHANGELOG.md)
CVS files changed	2 (1 corrected + 1 new CHANGELOG.md)
Canon docs updated	None this pass
Kernel	Unchanged — SHA-256: 7b08c024b77a24830c15e7952d6e54bed383aa960f4c74a71ff95ce51f4d80f5
Seal status	Pre-seal. This pass does not affect the seal target.

1. Executive Summary

This report documents the hardening pass conducted on 17 April 2026 across both the 512 repository (JonathanMastersWatson/512) and the Evidence-Sidecar CVS repository (JonathanMastersWatson/Evidence-Sidecar).

This pass addressed a single, specific, non-negotiable architectural error: GAP was being attributed to 512 — the Commit Gate — as a third gate output alongside ALLOW and DENY. This is incorrect.

The correct separation is:

- 512 produces exactly two outputs: ALLOW or DENY
- CVS records the ALLOW or DENY result from 512, and separately classifies any ungoverned execution periods as evidence chain gaps

GAP is a CVS witness layer classification applied to ungoverned periods in the evidence chain. It is not a gate output under any operational state — including pre-enforcement evaluation mode, observation mode, or fail-open. When the gate cannot evaluate, it produces no output. The fail-open handler engages, execution proceeds, and the witness layer records the ungoverned period as an evidence chain gap.

6 files in the 512 repo and 1 file in the CVS repo contained this error. All have been corrected. CHANGELOG.md was created in both repos to record this pass.

2. The Error — What Was Wrong and Why It Matters

Across multiple operational reference files, GAP appeared in gate output lists alongside ALLOW and DENY:

```
Result: ALLOW / DENY / GAP          ← incorrect
Valid outputs are exactly three...  ← incorrect
GAP results — evaluation failures   ← incorrect framing
| ← ALLOW/DENY/GAP — |             ← incorrect schematic
```

This framing is structurally incorrect for two reasons.

First, 512 is binary by definition. The gate evaluates all seven invariants and returns ALLOW if all pass, or DENY if any fail. Those are the only two states the gate can return. A gate that returns three states is not a Commit Gate — it is a policy engine.

Second, GAP is a witness layer concept. It describes a period in the evidence chain during which no gate evaluation occurred — because the gate was unavailable (fail-open). The gap is what the witness layer records about the ungoverned period. It is not what the gate returns. The gate returned nothing, because it was unavailable.

The distinction is not semantic. A system that believes its gate produces three outputs will build its integration, evidence model, and auditability claims around a false architectural premise. That premise does not survive regulatory scrutiny.

Note: the Gap Semantics section in ANTI_DRIFT.md (512-main) was already correct — it read: "A gap is not an evaluation result. It is a failure of evaluation." The surrounding sections contradicted this. Those contradictions are what this pass corrects.

3. Correct Architecture — The Separation Established

The correct model, now consistently expressed across both repos:

Layer	What it produces
Commit Gate (512)	ALLOW or DENY — binary, always, no exceptions
CVS Witness Layer	Records ALLOW or DENY result from gate evaluation. On fail-open events: records the ungoverned period as an evidence chain gap.

Fail-open behaviour — the corrected model:

- Gate is unavailable or evaluation times out
- Gate produces no output — evaluation did not complete
- Fail-open handler engages: commit path opens, execution proceeds
- Fail-open handler emits a gap record to the witness layer
- Witness layer classifies the ungoverned period as an evidence chain gap
- The gap is recorded, anchored, and must not be concealed

The schematic — corrected:

Normal path:

```
Proposed Action -> [512 GATE] -> ALLOW or DENY -> Execution
                        |
                        CVS records result
```

Fail-open path:

```
Proposed Action -> [512 GATE UNAVAILABLE] -> Execution proceeds
                        |
                        [fail-open handler]
                        |
                        gap_record -> CVS
                        CVS records ungoverned period as evidence chain gap
```

4. What Changed — 512 Repository

6 files corrected. 1 new file created.

File	Action	Changes
ANTI_DRIFT.md	Corrected	4 instances: gate output list, observation mode results list, advisory operation output count (three → two), and wording of advisory output rule.
512-ops/COMMIT_BOUNDARY_REFERENCE.md	Corrected	\$4 heading (three outputs → gate outputs), \$4 GAP subsection rewritten as Fail- Open (Gate Unavailable), \$5 XRPL table GAP row → Fail-open (gap record), \$5 intro corrected, \$7 Quick Reference table outputs row and GAP row corrected.
512-ops/REFERENCE_FLOW.md	Corrected	Stage 3 GAP- as-result sentence corrected. Stage 4 heading (three → two values), GAP subsection replaced with Fail-Open (Gate Unavailable). Evidence Object schema overall_result corrected. Sequence diagram gate signal corrected (ALLOW or DENY). Fail- open path arrow removed (gate produces no signal when unavailable).

File	Action	Changes
512-ops/INTEGRATION_STEPS.md	Corrected	Step 5 results list corrected. GAP results bullet reframed as Fail-open events. Step 5 output line and proceed condition corrected. Step 6 GAP sentence corrected. Step 7 GAP records reframed as witness layer gap records.
512-ops/PROPERTIES_CHECKLIST.md	Corrected	Evaluation Properties: three output values → two output values. Post-enforcement verification: GAP records → witness layer gap records.
USE_CASES/ENTRY_POINTS/ENTERPRISE_PRACTITIONERS.md	Corrected	Observation mode results list corrected. GAP results bullet reframed as Fail-open events.
CHANGELOG.md	New file	Creates repository changelog. Records this hardening pass as first entry, preserving the 2025-12-26 initial archive freeze as founding entry.

5. What Changed — Evidence-Sidecar (CVS) Repository

1 file corrected. 1 new file created.

File	Action	Changes
ANTI_DRIFT.md	Corrected	§1.1 Commit Gate (Enforcement Layer): 'Produces binary output only: allow / deny / gap' → 'Produces binary output only: ALLOW or DENY'. Single instance. All other sections correctly scoped and unchanged.
CHANGELOG.md	New file	Creates repository changelog for Evidence-Sidecar-main. First entry records this hardening pass. Notes coordination with parallel pass across six files in 512-main.

6. What Was Left Unchanged — And Why

The following were confirmed clean and required no changes:

Item	Reason unchanged
Gap Semantics section — ANTI_DRIFT.md (512)	Already correct: 'A gap is not an evaluation result. It is a failure of evaluation.' Surrounding sections contradicted this; those were corrected. This section was not touched.
Fail-Open Properties checklist — PROPERTIES_CHECKLIST.md	Gap record generation, forwarding, and identification items are correctly scoped to the witness layer. No change required.
README.md — both repos	No GAP-as-gate-output errors found. Confirmed clean.
02_EVIDENCE_MODEL/DELETION_AND_GAPS.md — CVS	GAP references correctly scoped to evidence chain classification. No change required.
Canon documents (512_ARCHITECTURE, CVS_ARCHITECTURE, etc.)	Not in scope for this pass. No changes made.
Kernel, invariants, XRPL anchor	Unchanged. This pass touches documentation only.

7. Seal Status

Kernel	Frozen and unchanged since genesis. SHA-256: 7b08c024b77a24830c15e7952d6e54bed383aa960f4c74a71ff95ce51f4d80f5
Invariants	Unchanged. 7 invariants. No additions, removals, or modifications.
XRPL anchor	Unchanged. Transaction: 378536A3CB75DECF90B6AE57F75292BDFF716285B01946870CAC158F8152D100
512 repo seal hashes	DF27F6C5C8DDBBD5341FB15EA943D92B3388331B386C26F44A145673F6C8D218 (2026-03-24) — to be recomputed after this pass
CVS repo seal hashes	0DA1ABA3A7257B636C0A364920C5CE643843257D217C0DBD68BA5DB64712BE17 (2026-03-24) — to be recomputed after this pass
Canon docs	Not updated this pass. Deferred.
CANON_HASHES.md	Not resealed this pass. Held until after next repo hardening pass.
Impact on seal target	This pass is terminology correction only. No structural or content changes to canonical documents. Seal target unaffected.

8. What Remains

Item	Notes
CVS canon document hardening pass	CVS_ARCHITECTURE and CVS_IMPLEMENTATION full hardening pass deferred. Next major pass.
512_ARCHITECTURE and 512_IMPLEMENTATION updates	Canon document updates identified in prior pass. Still deferred.
CANON_HASHES.md resealing	Held until after next full repo hardening pass on both repos.
REFERENCE_FLOW.md completion	Placeholder sections remain. Deferred.
Legal review package	Four placeholder files for counsel briefing. Deferred.
Constraint-Architecture repo pass	Not touched this pass.
Cross-document version reference audit	Full audit of version references across canon documents. Deferred.

9. Closing Statement

This pass is narrow in scope and exact in execution. One architectural error — GAP attributed to the Commit Gate as a third output — appeared across 7 files in two repositories. Every instance has been corrected. The gate produces ALLOW or DENY. The witness layer classifies ungoverned periods as evidence chain gaps. These are different systems with different functions. The documentation now reflects that consistently.

No kernel changes. No invariant changes. No canon document changes. The canonical SHA-256 hash is identical to the genesis commitment. This pass corrects the record.

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